Panskura Banamali College (Autonomous)

B.Sc.(H)6th Semester Examination -2022 (Online Mode)

Subject: Computer Science Paper:DSE3P

Paper Name: Machine LearningLab
F.M.:20 Time:2Hours

Instruction

[Make your answer script into PDF and MP4 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020UG009967, paper_code : DSE3P, then file name will be :: 2020UG009967- DSE3P.pdf for question 'A' and 2020UG009967- DSE3P.mp4 for question 'B']

A. Answer any one question: **1 x12 = 12**

[Using: Octave/MATLAB/python/Jupiter lab/Jupiter note book/Colab]

1. Perform elementary mathematical operations in like addition, multiplication, division and exponentiation.
2. Create various type of plots/charts like histograms, plot based on sine/cosine function based on data from a matrix. Further label different axes in a plot and data in a plot.
3. Perform basic operations on matrices (like addition, subtraction, multiplication) and display specific rows or columns of the matrix.
4. Perform vectorized implementation of simple matrix operation like finding the transpose of a matrix, adding, subtracting or multiplying two matrices?
5. Implement Linear Regression problem to draw a plot year vs. salary. For example, based on a dataset comprising of existing set of prices and area/size of the houses, predict the estimated price of a given house.
6. using multiple linear regression to predict prices and find out the score, the score must be greater than 95%.

Marks distribution:

a. Problem Definition -	1 Marks
b. Algorithm/Flow Chart/ Pseudo-code–	5 Marks
c. Program listing -	3 Marks
d. Output/Result with discussion-	3 Marks

B. Answer any four questions: **4 x2 = 8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is Machine Learning?
2. What are the different types of Machine Learning?
3. Why do we need Machine Learning?
4. what is supervised machine learning ?
5. Which assessment is used to test the intelligence of the machine?
6. What is Bias in machine learning ?
7. what is classification?
8. what is regression?

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